# TraceErase: AI-Powered Platform to Track and Prove Personal Data Deletion

Oct 21, 2026 · @Sanchay

## 1. Overview

TraceErase finds every copy of your personal data, deletes it everywhere with one click, and proves which companies actually deleted it.

| Item | Detail |
|---|---|
| Project name | TraceErase |
| Tagline | Apps say "deleted". TraceErase proves it, everywhere your data went. |
| Hackathon | DEXATHON 2026, Sathyabama Institute of Science and Technology, 24-hour AI hackathon, 4–5 November 2026 |
| Track | Blockchain & Cybersecurity |
| Problem statement | Problem 4: Verifying whether personal data has actually been deleted |
| Type of product | Web application, installable on phones as a PWA, plus a browser extension and a free connector for companies |
| Who it helps | Every internet user, companies that want to prove compliance, and privacy regulators |

**In one line:** TraceErase gives every person a protected identity, a map of where their data went, one-click deletion everywhere, and proof of who obeyed and who lied, whether the company cooperates or not.

**Note on examples:** real apps (Zomato, Razorpay, Myntra, MakeMyTrip, Flipkart) are used only to make the story easy to follow. Which of them use the connector, and who they share data with, is imagined for this document. It does not describe their actual practices. PizzaPoint is a fictional company.

# 2. The problem

When you delete your data from an app, it does not really disappear everywhere, and you have no way to check.

## 2.1 The problem in simple words

You sign up on a shopping app and give it your name, phone number, email and address. Without telling you clearly, the app shares your details with other companies:

- a delivery company, to deliver your order
- an advertising company, to show you ads
- an analytics company, to study what you buy
- a backup server, to keep copies safe

Your data is now in five places, not one. Later you click "Delete my account", and the app says "Your data has been deleted". In reality, only the app deleted it. The delivery company, the ad company, the analytics company and the backup server still have copies. That is why people keep getting marketing calls and messages months after deleting an account.

**Everyday comparison:** it is like giving photocopies of your Aadhaar card to five shops. Later you ask one shop to tear up its copy, and it does. The other four still have theirs, and you cannot check whether they threw them away.

## 2.2 Key pain points

1. **You do not know where your data went.** It travels to companies you never dealt with directly.
2. **Deleting from one app does not delete the copies.** Partner companies, caches, backups and derived datasets keep them.
3. **Every company handles deletion differently.** Some ignore requests, some delay, some delete only partly.
4. **A "deleted" message is not proof.** Nothing independently confirms that every copy is gone.
5. **People have no visibility** into what happens to their data after sharing it.

## 2.3 Why this problem matters now

Privacy laws already give people the right to have their data erased:

| Law | Where | What it gives users |
|---|---|---|
| Digital Personal Data Protection (DPDP) Act, 2023 | India | Right to erasure; penalties up to ₹250 crore for violations |
| GDPR, Article 17 | European Union | Right to erasure ("right to be forgotten"); fines up to 4% of global turnover |
| CCPA | California, USA | Right to delete personal information |

The laws exist, but there is no easy way for an ordinary person to check whether companies actually obeyed them.

### 2.4 The problem in one line

Apps say your data is deleted, but copies remain with other companies, and you have no way to know or prove it.

## 3. The solution

TraceErase protects your identity when you sign up, tracks where your data travels, deletes it everywhere with one click, proves who deleted it, and catches companies that lie.

### 3.1 The six things TraceErase does

| # | Step | What it means in simple words |
|---|---|---|
| 1 | **Protect** | You sign up on apps with a protected email and phone number instead of your real ones. Messages are forwarded to you, but your real details stay hidden. |
| 2 | **Track** | Companies that use the TraceErase Connector send a receipt every time they share your data. Each receipt is fingerprinted on a blockchain so nobody can change it. |
| 3 | **Discover** | AI finds the companies that already have your data (old apps) and reads their privacy policies to learn who they share with and how long they keep data. |

| # | Step | What it means in simple words |
|---|---|---|
| 4 | **Delete** | One click sends deletion requests to every company: automatically to connected companies, and through AI-written legal requests to the rest. |
| 5 | **Prove** | Connected companies send signed deletion receipts. Others must send an official email confirmation, which is verified and fingerprinted. You get a deletion certificate. |
| 6 | **Catch** | Each protected email works as a trap (a canary). If an unknown company uses it, that proves your data was shared or not deleted. TraceErase blocks the leaker, demands deletion and prepares a complaint. |

## 3.2 The three parts of the system

TraceErase is a bridge between users and companies:

- **User side: the TraceErase app.** A web application that installs on phones as a PWA, plus a browser extension for laptops.
- **The bridge: a shared ledger.** A blockchain that holds only fingerprints and receipts, plus an encrypted database that holds the readable details, visible only to the user.
- **Company side: the TraceErase Connector.** Free, open-source software that companies add to their own servers. It sends receipts when they share or delete data.

## 3.3 Two modes, so it works even if companies do not join

- **Mode A, connected companies:** full tracking, automatic deletion, signed receipts, blockchain proof.
- **Mode B, non-connected companies:** protected emails and phones catch leaks, AI sends legal access and deletion requests, AI checks replies, and complaints go to the regulator.

The system works even if zero companies join. Companies that join simply give stronger proof.

### 3.4 Who uses TraceErase

| User | What they get |
|---|---|
| Everyday people | A protected identity, a map of where their data went, one-click deletion, proof, leak alerts |
| Companies | An easy way to record sharing, automate deletion and prove they follow the law |
| Regulators, consumer groups, journalists | A transparency board and dashboard showing which companies ignore deletion requests |

## 4. Key concepts explained simply

Eight ideas make TraceErase work; each is explained here in plain words before the full journey.

### 4.1 Protected email

A separate email address that TraceErase creates for each app, for example `sanchay.zomato.k7x9@traceerase.app`.

- You use it instead of your real email when you sign up.
- Emails sent to it arrive at TraceErase's server first. TraceErase checks the sender, then forwards good emails to your real inbox.
- Email content is not stored. Only the sender name is logged.
- TraceErase does not need to read your personal Gmail for this.

### 4.2 Protected phone number

One virtual phone number per user, for example +91 70000 12345. Calls, SMS and WhatsApp messages to it are forwarded to your real phone, and TraceErase logs who contacted you.

- One number per user (not per app), because virtual numbers cost money. Per-app numbers can be a premium option.
- Banks, UPI and government services often need your real SIM number. You keep using your real number for those.

### 4.3 Canary (the trap)

Every protected email is given to only one app. If any other company sends a message to it, that proves the app shared your data, or did not delete it. The protected phone number works the same way. The name comes from the old practice of miners carrying canaries to detect danger early.

### 4.4 Allowed senders list

For each protected email, TraceErase keeps a list of who is allowed to use it: the app itself, plus partners shown in its sharing receipts. Anyone else is flagged as a leak. This allows smart blocking: block only the leaker, not the whole email.

### 4.5 TraceErase Connector

Free, open-source software that a company adds to its own servers, like adding Razorpay payments or Google login.

- **Simple comparison:** a supermarket billing machine prints a receipt automatically every time a sale happens. The connector is a receipt printer for data sharing.
- When the company's system shares a customer's data with a partner, its own code also sends a receipt to TraceErase. This is what "the system calls the connector" means.
- It is not hardware. It never enters or reads the company's database. It only sends receipts when the company's system triggers it.

### 4.6 Blockchain (the notebook nobody can erase)

A shared digital record that no single party can change or delete, not even TraceErase. It is used because many companies are involved and none of them fully trust each other.

- Only fingerprints (hashes), signatures and timestamps go on the blockchain. No personal data ever goes there, because a blockchain can never delete anything.
- **Bundling:** thousands of receipts are combined into one fingerprint (a Merkle root) and written once an hour. This makes the cost almost zero, and any single receipt can still be proven to be inside the bundle.

### 4.7 Passkey (your digital key)

A secret digital key created on your phone and unlocked with your fingerprint. Every deletion request is signed with it, so companies know the request really came from you. The key never leaves your phone.

### 4.8 DKIM (proof inside an email)

Company email servers automatically add a digital signature called DKIM to the emails they send. It proves the email really came from that company's domain and was not changed. TraceErase uses it to turn a company's "we deleted your data" email into tamper-proof evidence.

### 4.9 Deletion statuses

| Status | Meaning |
|---|---|
| Verified deleted | Signed receipt from a connected company, fingerprinted on the blockchain |
| Partially deleted | Some data kept for a legal reason, such as tax invoices; the reason is shown |
| Claimed deleted | Official email confirmation from a non-connected company, DKIM-verified, with the canary still watching |
| Pending | Request sent, still within the deadline |
| No response | Deadline passed; AI is following up and escalating |
| Overdue | A declared retention period ended without a deletion confirmation |
| Leak detected | The canary caught an unknown company using a protected email or number |

## 5. The complete user journey (Sanchay's story)

This section follows Sanchay from installing TraceErase to receiving his deletion certificate, covering both connected and non-connected companies.

### 5.0 The companies in this story

| Company | What it is | Sanchay's situation | Connected to TraceErase? |
|---|---|---|---|
| Zomato | Food delivery | New sign-up | Yes (imagined) |
| Razorpay | Zomato's payment partner | Receives data from Zomato | Yes (imagined) |

| Company | What it is | Sanchay's situation | Connected to TraceErase? |
|---|---|---|---|
| An ad network | Zomato's advertising partner | Receives data from Zomato | No |
| Myntra | Shopping | New sign-up | No |
| MakeMyTrip | Travel | Old account, real email, 3 years ago | No |
| Flipkart | Shopping | Old account, "Sign in with Google" | No |
| PizzaPoint | Unknown company (fictional) | Secretly gets his data: the leak | No |

## Phase 1: Installing TraceErase and setting it up

1. **Open the website.** Sanchay opens Chrome on his phone and goes to traceerase.app.
2. **Install it as an app (PWA).** Chrome shows "Install TraceErase?" He taps Install. TraceErase now has a home-screen icon and opens full-screen like a normal app. The same web app works on his laptop, with the same account, and updates instantly without the Play Store.
3. **Add the browser extension (laptop).** It fills in protected emails on websites and fills company request forms.
4. **Create an account.** He enters his real email (sanchay@gmail.com) and real phone (98765 xxxxx), verified by OTP. These are the only personal details TraceErase stores, encrypted, used only to forward messages to him.
5. **Create a passkey.** He confirms with his fingerprint. A secret digital key is created on his phone. Every deletion request will be signed with it. The key never leaves his phone.
6. **Get a protected phone number.** TraceErase gives him +91 70000 12345. Calls, SMS and WhatsApp to it are forwarded to his real phone. He keeps his real number for banks, UPI and government services.
7. **Allow notifications.** TraceErase can now alert him about receipts, deadlines and leaks.
8. **See the safety promise.** Minimum data stored; emails pass through and are never stored; AI never sees raw personal details; nothing personal on the blockchain; permissions can be turned off any time; TraceErase itself can be deleted, with its own signed receipt.

## Phase 2: Finding old apps (cleaning up the past)

**Sources Sanchay allows:**

| Source | How it works |
|---|---|
| Gmail, read-only | Connected through Google's official permission screen |
| "Sign in with Google" apps | The list of apps connected to his Google account |
| Manual add | Any app he remembers |

**What the AI does:** it looks at welcome emails, order receipts, OTP emails and newsletters to find company names. It keeps only the company names and discards everything else. No email content is stored.

**Result:** "We found 41 companies with your data. 12 you haven't used in over a year." These include MakeMyTrip (real email, 3 years ago) and Flipkart ("Sign in with Google").

**AI reads their privacy policies.** For each company, the AI reads its public privacy policy and the Data Safety section of its Play Store listing. It extracts what data is collected, who it is shared with, how long it is kept, and how to request deletion (grievance officer contact, forms). Example: "MakeMyTrip shares with airlines, hotels and marketing partners. Marketing partners are not named. Retention is not clearly stated."

## Phase 3: Signing up on a connected company (Zomato)

1. **Get a protected email.** Before signing up, Sanchay opens TraceErase, taps "+ New sign-up" and types Zomato. TraceErase creates `sanchay.zomato.k7x9@traceerase.app` (copied automatically) and shows his protected phone number.
2. **Sign up.** On his phone he pastes the protected email and number into Zomato (about 3 seconds). On a laptop, the extension fills them in automatically.
3. **Zomato never gets his real email or real phone.** He orders biryani.
4. **Allowed senders list.** TraceErase notes that this email belongs to Zomato. Partners are added automatically as receipts arrive. Anyone else emailing this address is a warning sign.

## Phase 4: What the connector does inside Zomato

**What "the system calls the connector" means:** Zomato's server has a function that sends a customer's payment details to Razorpay. When Zomato joined TraceErase, its developers added one small piece of code there: "Whenever we share a customer's data with a partner, also send a receipt to TraceErase." So Zomato's own computer sends a receipt automatically every time it shares data. No person does anything.

**Receipt 1, 8:02 PM, Zomato to Razorpay (payment):**

| Field | Value |
|---|---|
| User | a7f3… (a secret code, not his name) |
| From → To | Zomato → Razorpay |
| Data shared | Name, protected phone, payment reference |
| Purpose | Process payment |
| Keep until | As required by financial law |
| Data fingerprint | 0x8a2f… (proves what was sent, without showing it) |
| Signed by | Zomato |

**Receipt 2, 8:05 PM, Zomato to the ad network (advertising):** data shared is the protected email, food preferences and area; purpose is personalised ads; keep until 12 months; signed by Zomato.

**Who decides "keep until 12 months"?** The company decides, and the law requires it to: data may be kept only as long as needed for its purpose, and users must be told. The connector takes this value from the company's own retention policy. Because the receipt is signed, it is an official promise the company cannot deny later.

**Where receipts go:** full readable details go to TraceErase's encrypted database, visible only to Sanchay. Fingerprints are bundled hourly into one fingerprint written to the blockchain, at a fraction of a rupee per batch.

**The ad network is not connected, but we still know.** The ad network never installed the connector. But Zomato's connector recorded that Zomato sent data to it. So TraceErase knows the ad network has Sanchay's data, and adds it to the allowed senders list for the Zomato protected email.

## Phase 5: Signing up on a non-connected company (Myntra)

Sanchay taps "+ New sign-up", types Myntra, and gets `sanchay.myntra.p2m4@traceerase.app`. He pastes it, with his protected phone, into Myntra.

Myntra has no connector, so no receipts arrive. TraceErase still protects him:

| Method | What it does |
|---|---|
| AI reads the privacy policy | "Shares with payment, logistics and marketing partners. Keeps data until account deletion plus a legal period." |
| Canary | Any sender other than Myntra using this email proves Myntra shared his data |
| Legal access request (optional) | AI asks Myntra what data it holds and who it shared it with; Myntra must legally reply |

**How the AI contacts Myntra (in this order):**

1. **Email to the grievance or privacy officer.** The contact is published in Myntra's privacy policy. The email is sent from `sanchay.myntra.p2m4@…`, the email registered on his account, so Myntra knows it comes from the real owner.
2. **Myntra's own request form.** Filled automatically by the extension on a laptop, or guided step by step on a phone.
3. **Formal letter.** The last option, prepared by AI.

## Phase 6: Daily life (TraceErase working quietly)

**Email flow:** Zomato sends "Your order is on the way". It arrives at the Zomato protected email on TraceErase's server. The sender is on the allowed list, so it is forwarded to his Gmail. The content is not stored.

**Phone flow:** the delivery partner calls the protected number and the call is forwarded. OTP SMS messages are forwarded instantly. Only the sender is logged.

**Data map:** Sanchay sees Zomato (connected) with Razorpay (connected) and the ad network (not connected) under it; Myntra with partner categories from its policy; MakeMyTrip and Flipkart as old accounts; and 37 more. The AI explains: "Zomato shared your food preferences with an ad network for 12 months. It is not connected, so we cannot see what it does next, but your protected email will catch any misuse."

**Retention timers:** for every receipt, TraceErase sets a timer. The ad network's timer ends at month 12 (see Phase 10).

## Phase 7: Fixing old accounts

**MakeMyTrip (old account with real email):**

1. **Switch email.** TraceErase creates `sanchay.makemytrip.r5t1@…`. The AI guides him: My Profile, Edit email, paste, verify OTP. From now on, the canary watches MakeMyTrip.

2. **Ask what they have.** The AI sends a legal data-access request to MakeMyTrip's grievance officer: "Under the DPDP Act, tell me what data you hold about me and name every company you shared it with."
3. **Read the reply.** The AI reads MakeMyTrip's reply and adds the named partners to his map.
4. **Mark for deletion.** He no longer uses it, so he adds it to the deletion list.

**Honest limitation:** MakeMyTrip had his real email for 3 years. If it leaked during that time, the canary cannot prove it, because it only catches leaks after the switch. That is why the legal access request matters for old accounts.

**Flipkart (old account with "Sign in with Google"):**

1. **Cut the Google link.** The AI guides him to Google Account, Security, Third-party apps with account access, Flipkart, Remove access. This stops Flipkart getting new data from Google.
2. **Protected phone.** The AI suggests updating Flipkart's phone number to his protected number, so the phone canary can watch.
3. **Request deletion.** The AI sends a legal deletion request to Flipkart's grievance officer.

## Phase 8: "Delete my data everywhere"

Sanchay selects Zomato (and its partners), Myntra, MakeMyTrip and Flipkart, and taps "Delete my data everywhere". He confirms with his fingerprint. This does two things:

1. **Signs** the request with his passkey, proving it is really him.
2. **Pre-authorises** TraceErase to complete identity checks, such as OTPs, only for these deletion requests.

**8A. Zomato (connected): fully automatic**

| Step | What happens | Who does it |
|---|---|---|
| 1. Identity check | Is the request signed by the real owner of the account linked to the Zomato protected email? Yes. | Zomato's system |
| 2. Legal hold check | Is there data that must be kept by law? Tax invoices are kept; everything else is deleted. | Zomato's system |
| 3. Delete | Profile, addresses, order history and preferences are erased. | Zomato's system |
| 4. Signed receipt | "Zomato deleted user a7f3… at 10:14 AM. Kept: tax invoice (legal reason)." | Connector |

| Step | What happens | Who does it |
|---|---|---|
| 5. Forward to partners | Zomato's receipts show sharing with Razorpay and the ad network, so the request is forwarded to both. | Connector |

This finishes in minutes with no human involved.

**8B. Razorpay (connected partner): automatic.** Identity is verified because the request came signed via Zomato. Payment records must be kept by financial law, so they are kept; contact details are deleted. A signed receipt is sent.

**8C. The ad network (not connected): semi-automatic.** The AI sends a formal deletion notice to its privacy contact, attaching Zomato's signed sharing receipt as proof that it received the data. A deadline timer starts.

**8D. Myntra (not connected): semi-automatic, with OTP handled automatically**

1. **Send the request.** The AI emails Myntra's grievance officer from `sanchay.myntra.p2m4@…`, or fills its request form.
2. **Identity check by OTP.** Myntra sends an OTP to the protected email or phone. Both arrive at TraceErase's server first, so TraceErase receives the OTP and the AI submits it within seconds, even if Sanchay is asleep. This is allowed only because he pre-authorised this deletion. It never works for logins or payments, and every use is logged and shown to him.
3. **Backup.** If the OTP fails or expires, the AI requests a new one and re-sends the request. A company cannot legally ignore a deletion request just because an OTP expired.
4. **Myntra processes it.** Myntra's team deletes the data manually.
5. **Myntra replies.** The AI checks the reply for the required details.

**What a proper confirmation must contain:**

| Must include | Example |
|---|---|
| What data was deleted | Name, email, phone, addresses, orders |
| Date of deletion | 2 Nov 2026 |
| What was kept, and why | Invoices kept for tax law |
| Partners informed | "Logistics and marketing partners asked to delete" |
| Reference number | #MYN-48213 |
| Official sender | Grievance or privacy officer |

If the reply is vague, such as "Your account is closed", the AI automatically replies asking for the missing details. The confirmation email carries a DKIM signature proving it came from Myntra and was not changed. TraceErase stores the email's fingerprint on the blockchain. The status becomes **Claimed deleted**: an official, signed confirmation stored as proof, with the canary still watching.

**8E. MakeMyTrip and Flipkart (not connected).** The same as Myntra. The AI sends the requests, also asking MakeMyTrip to delete from the partners it named. It handles OTPs automatically, checks the replies and starts deadline timers.

## Phase 9: Results dashboard and follow-ups

**Sanchay's dashboard on day 10:**

| Company | Connected? | Status |
|---|---|---|
| Zomato | Yes | Partially deleted: tax invoice kept by law |
| Razorpay | Yes | Partially deleted: payment record kept by law |
| Ad network | No | No response |
| Myntra | No | Claimed deleted, canary watching |
| MakeMyTrip | No | Claimed deleted, canary watching since the switch |
| Flipkart | No | Pending |

**Automatic follow-up for the ad network:**

| Day | Action |
|---|---|
| Day 7 | AI sends a reminder |
| Day 10 | AI files a formal grievance with the ad network (the law requires this before going to the regulator) |
| Day 15 | Still nothing: the complaint is prepared automatically with all evidence |

Sanchay gets a notification: "Your complaint against the ad network is ready. Review and submit?" He taps Submit. It is one tap, not fully automatic, because a legal complaint is filed in his name and he must agree to it. An optional setting, "Auto-file complaints for me when deadlines pass", makes it fully automatic if he chooses.

## Phase 10: Retention expiry (data he did not delete himself)

If Sanchay keeps using an app, his data is still protected by the retention timers. Example: Zomato shared data with the ad network for 12 months.

1. At month 12, the timer ends.
2. A connected company's system deletes the data automatically, and the connector sends an expiry deletion receipt.
3. For a non-connected company, the AI sends: "Your declared period has ended. Please delete and confirm."
4. No confirmation means the status becomes **Overdue**, and the grievance and complaint flow starts.
5. The canary keeps watching. Any use after month 12 is proof the data was not deleted.

TraceErase cannot look inside companies. It relies on signed promises, timers, follow-ups and canary checks. A broken signed promise becomes evidence.

## Phase 11: The canary catches a leak

**The leak.** On day 40 an email arrives: "Flat 50% off at PizzaPoint". It was sent to the Zomato protected email. PizzaPoint is not on the allowed list, which contains only Zomato, Razorpay and the ad network.

**AI investigation:** "This address was given only to Zomato. Zomato's receipts show it was shared with Razorpay (deleted) and the ad network (never responded). PizzaPoint is not in any record. Most likely the ad network passed your data to PizzaPoint, after you asked for deletion."

**Smart blocking (block the leaker, not the email):**

| Sender | Result |
|---|---|
| Zomato | Still forwarded, if he still uses Zomato |
| PizzaPoint | Blocked |
| Any other unknown sender | Blocked automatically in optional strict mode |

If leaks keep happening, TraceErase **rotates** the email: it creates a new Zomato protected email, guides him to update it in Zomato, and shuts the old one down, so every leaker loses access at once. His real email was never exposed.

**Deleting the data PizzaPoint holds.** Blocking stops contact, but PizzaPoint still has the data. So the AI:

1. Adds PizzaPoint to his map as a "leaked receiver".

2. Sends a legal access request: "Where did you get my data?"
3. Sends a deletion request with the canary evidence attached.
4. Tracks it through confirmation, follow-up, grievance and complaint.

**Evidence pack (built automatically):**

| Evidence | Source |
|---|---|
| Zomato shared data with the ad network on day 1 | Blockchain receipt, signed by Zomato |
| Deletion requested on day 31 | Blockchain record |
| No receipt from the ad network | Dashboard record |
| Reminder and grievance ignored | Email log, DKIM-verified |
| PizzaPoint emailed the protected address on day 40 | Canary log |

## Phase 12: Complaint, regulator and transparency board

**Complaint.** A pre-filled complaint for the Data Protection Board of India, the government authority under the DPDP Act, covers the ad network (which leaked the data) and PizzaPoint (if it does not delete). Sanchay submits it through the Board's official complaint channel. The Board can order deletion and impose penalties of up to ₹250 crore.

**Public transparency board (live now).** A public, anonymised board shows, for example: "Ad network: 1,247 ignored deletion requests, 312 leaks detected this month." Journalists, consumer groups and NGOs can use it today.

**Regulator dashboard.** It is built and working now. Official use by the Data Protection Board needs a government partnership. That is paperwork, not technology.

### Phase 13: The deletion certificate

```
TRACEERASE DELETION CERTIFICATE
User: #a7f3…        Request date: Day 31

Zomato       Partial: tax invoice kept by law     (Receipt 0x8a2f…)
Razorpay     Partial: payment record kept by law  (Receipt 0x3c91…)
Myntra       Claimed deleted (DKIM-verified, no leaks)
MakeMyTrip   Claimed deleted (DKIM-verified)
Flipkart     Pending
Ad network   Not deleted -> complaint filed (Ref #DPB-xxxx)
PizzaPoint   Leaked receiver -> blocked + deletion requested

Receipts verifiable on blockchain - cannot be altered
```

**What happens after the certificate:** the certificate itself does not delete anything. It is proof, and proof forces action.

| Situation | Outcome |
|---|---|
| Verified or partial deletion | Solved, with proof forever |
| Claimed deleted | The canary keeps watching; any leak becomes new evidence |
| Ignored request | Complaint to the regulator, which can order deletion and fine the company |
| Leaked receiver | Blocked instantly, deletion demanded, included in the complaint |

### Phase 14: Leaving TraceErase

In Settings, Sanchay taps "Delete my TraceErase account". His real email, phone, protected addresses and records are erased, and he receives TraceErase's own signed deletion receipt. TraceErase practises what it preaches.

## 6. Connected vs non-connected companies

Connected companies give automatic deletion and signed proof; non-connected companies are still covered, through legal requests and canaries, with weaker proof.

| Step | Connected (Zomato, Razorpay) | Not connected (ad network, Myntra, MakeMyTrip, Flipkart) |
|---|---|---|
| Knowing who got the data | Automatic receipts from the connector | Privacy policy, legal access request, and the sharing company's receipts if the sharer is connected |
| Retention time | Declared and signed in each receipt | AI reads it from the privacy policy, or asks the company |
| Leak detection | Receipts plus canary | Canary (protected email or phone) |
| Sending the deletion request | Automatic through the connector | Email to the grievance officer, the company's form, or a formal letter, written by AI |
| Identity check | Passkey signature, automatic | OTP received by TraceErase and submitted automatically |
| Deletion | Automatic, in minutes | Company staff, in days |
| Proof | Signed receipt fingerprinted on the blockchain | DKIM-signed email confirmation fingerprinted on the blockchain, plus canary monitoring |
| If ignored | Rare; flagged automatically | Reminder, grievance, complaint (one tap or auto-file) |

**Key insight:** even when a partner is not connected, the company that shared the data with it can be. Its connector records the share, so TraceErase knows the partner has the data even though the partner never said so.

## 7. Features by portal

TraceErase has three portals: users, companies, and regulators or the public. Every feature below uses AI, blockchain, or both.

### 7.1 User portal (PWA on phone and laptop, plus browser extension)

| # | Feature | What it does |
|---|---|---|
| 1 | Protected email per app | Creates a separate email for each sign-up and forwards good emails to the real inbox |

| # | Feature | What it does |
|---|---|---|
| 2 | Protected phone number | One virtual number per user; calls, SMS and WhatsApp forwarded |
| 3 | AI data discovery | Finds companies that already have your data from Gmail (read-only), the "Sign in with Google" list and manual entries |
| 4 | AI privacy policy reader | Reads policies and Play Store Data Safety labels: data collected, partners, retention, deletion contact |
| 5 | AI data map | Visual map of where your data travelled, explained in simple words |
| 6 | AI risk score | Flags companies holding sensitive data or with vague sharing policies |
| 7 | Old-account helper | Guides you to switch old accounts to protected emails, or to cut "Sign in with Google" access |
| 8 | One-click delete everywhere | Sends signed deletion requests to all selected companies and their partners |
| 9 | AI legal request writer | Writes access and deletion requests under DPDP, GDPR or CCPA, depending on where you live |
| 10 | AI auto-OTP handler | Completes identity checks for deletion requests you started, even while you sleep |
| 11 | AI reply checker | Reads company replies, checks they contain the required details, and asks for missing ones |
| 12 | AI follow-up agent | Reminders, grievances and escalation when deadlines pass |
| 13 | Retention timers | Track each company's declared "keep until" date and check for deletion afterwards |
| 14 | Canary leak detector | Flags any unknown sender using a protected email or number |
| 15 | Smart blocking and rotation | Blocks only the leaker, or replaces a leaked protected email with a new one |
| 16 | AI evidence pack and complaint | Builds the evidence and a pre-filled complaint for the regulator |

| # | Feature | What it does |
|---|---|---|
| 17 | Deletion certificate | Verifiable proof of what each company did, backed by blockchain receipts |
| 18 | Voice and chat assistant | Answers questions like "Has Zomato deleted my data?" |

## 7.2 Company portal (for apps and their partners)

| # | Feature | What it does |
|---|---|---|
| 19 | Connector (SDK / API) | Sends a receipt whenever the company shares or deletes a user's data; never reads its database |
| 20 | Automatic deletion forwarding | Passes a deletion request to every partner that received the data |
| 21 | Identity and legal hold checks | Verifies the user's passkey signature and keeps legally required data |
| 22 | Signed deletion and expiry receipts | Confirms deletion, including automatic deletion when a retention period ends |
| 23 | AI compliance assistant | Flags overdue requests and partners that are not complying |
| 24 | "TraceErase Verified" badge | Shows users the company proves its deletions |

## 7.3 Regulator and public portal

| # | Feature | What it does |
|---|---|---|
| 25 | Public transparency board | Anonymised ranking of companies that ignore deletion requests or leak data; usable now |
| 26 | AI non-compliance ranking | Finds repeat offenders across thousands of requests |
| 27 | AI complaint clustering | Groups many users' complaints against the same company |
| 28 | Independent audit trail | Anyone authorised can verify a deletion claim against the blockchain |

# 8. Where AI and blockchain are used, and why

Blockchain is used only for shared, tamper-proof records between parties who do not trust each other; AI is used wherever a human would otherwise have to read, write, chase or judge.

## 8.1 Why blockchain is genuinely needed

The organisers ask that blockchain be used only where shared verification is truly needed. TraceErase meets that test:

- Many different companies are involved, and none of them fully trust each other.
- No company should be able to later deny that it received the data.
- No company should be able to fake or edit a deletion receipt.
- Users and regulators need to verify records independently of TraceErase.

**What goes on the blockchain:** fingerprints (hashes) of sharing receipts, deletion receipts and DKIM-verified confirmation emails, plus timestamps and company signatures.

**What never goes on the blockchain:** names, emails, phone numbers, addresses or any personal data. A blockchain can never delete anything, so storing personal data there would defeat the purpose.

**Cost control:** receipts are bundled into one Merkle root per hour, so a single blockchain write covers thousands of receipts.

## 8.2 Where AI is used

| AI task | Why AI is needed |
|---|---|
| Finding old apps from emails | Thousands of emails must be sorted into company names |
| Reading privacy policies and Data Safety labels | Policies are long and confusing; AI extracts partners, retention and deletion contacts |
| Writing legal requests | Each country's law needs a different format and wording |
| Reading company replies | Replies vary; AI checks for required details and asks for missing ones |
| Handling OTPs | AI finds and submits the code in the right form or email |
| Detecting leaks | AI decides whether a sender is expected, a partner, or a leaker |

| AI task | Why AI is needed |
| --- | --- |
| Building evidence and complaints | AI assembles receipts, logs and timelines into a complaint |
| Spotting non-compliant companies | AI ranks repeat offenders and clusters complaints |
| Explaining in simple words | AI turns technical records into plain-language explanations |

## 9. Security and user trust

TraceErase is designed to know as little about you as possible, because a privacy tool that collects everyone's data would itself become a target for hackers.

| Principle | What it means |
|---|---|
| Minimum data | Stores only your real email and phone, encrypted, needed to forward messages |
| Emails pass through, never stored | Emails are forwarded and not saved; only the sender name is logged |
| Gmail scan, then forget | The AI keeps only company names from your inbox and discards everything else |
| Personal data hidden from AI | Names, numbers and addresses are masked before any AI reads text |
| Encryption | Stored data is encrypted; only you can see your own records |
| Nothing personal on the blockchain | Only fingerprints, so reading the blockchain reveals nothing about you |
| Passkey stays on your phone | Not even TraceErase can see or use your key |
| Auto-OTP is limited | Works only for deletion requests you started; never for logins or payments; every use is logged |
| Permissions you control | Gmail and other access are optional and can be turned off any time |
| Open-source code | Anyone can check that nothing hidden happens |
| Practise what we preach | You can delete your TraceErase account and receive a signed deletion receipt |

**How identity is checked.** Only Sanchay should be able to delete Sanchay's data. TraceErase uses three methods:

1. **Passkey signature (connected companies, automatic).** The deletion request is signed with the key on his phone, like a signature on a cheque that cannot be copied.
2. **OTP confirmation (non-connected companies).** The company sends an OTP to the protected email or phone. It arrives at TraceErase's server and is submitted automatically, or Sanchay taps Confirm.
3. **Login approval.** Some companies show a popup asking him to approve the request, like "Sign in with Google".

Because each account was created with a protected email, only TraceErase and Sanchay receive messages sent to it, which makes it very hard for anyone else to pretend to be him.

# 10. Cost, adoption and why companies will agree

TraceErase costs almost nothing to run, needs no hardware, and fits a role that India's privacy law already created.

## 10.1 Cost

| Who | Cost |
|---|---|
| Users | Free. Protected emails cost almost nothing. Extra per-app phone numbers can be a paid premium option. |
| Companies | The connector is free, open-source software. No hardware, no installation in their data centre. |
| Blockchain | Thousands of receipts bundled into one hourly write, a fraction of a rupee per batch |
| Infrastructure | Email forwarding and a database cost about the same as any normal web app |

## 10.2 The connector does not enter a company's systems

- It is software, added with a few lines of code, like Razorpay payments or Google login.
- It does not read the company's database. The company's own system calls it only when it shares or deletes data.
- It sends out only receipts and fingerprints, never actual customer data.
- Only the user sees their own data trail. Competitors see nothing, and partner names are mostly already in privacy policies.

## 10.3 We cannot force companies to join, and we do not need to

TraceErase works in two modes. Non-connected companies are still covered through protected emails, canaries, AI legal requests and complaints. The system works even if no company joins on day one.

## 10.4 Why companies will agree

1. **The law already requires deletion and disclosure.** Under the DPDP Act, penalties can reach ₹250 crore; under GDPR, fines can reach 4% of global turnover. Proof of compliance protects them.

2. **It saves money.** Handling deletion requests manually by email is slow and expensive; the connector automates it.
3. **It protects them in audits.** Signed receipts prove they obeyed the law.
4. **It builds trust.** A "TraceErase Verified" badge attracts privacy-conscious users.

### 10.5 An official path: the DPDP Consent Manager

The DPDP Act 2023 creates the role of a **Consent Manager**: an official platform, registered with the Data Protection Board, through which people manage and withdraw consent given to companies. TraceErase fits this role. It is not an outsider forcing its way in; it is the kind of platform the law already made room for.

**Precedent:** RBI's Account Aggregator framework created a regulated intermediary for sharing financial data between banks and apps. Banks adopted it because it was official, regulated and safe. TraceErase follows the same model for deletion.

**Answer for judges:** "TraceErase is free, open-source software, not hardware. It never touches company databases. It works even if companies do not join, through protected emails and legal requests. And it fits India's DPDP Consent Manager framework, so it can become an official, regulated platform, like the Account Aggregator system banks already use."

## 11. Frequently asked questions (all doubts answered)

These are the questions raised while designing TraceErase, with the answers judges should hear.

**Does TraceErase control the whole phone?** No. Phones do not let one app see inside another. TraceErase gives protected emails and numbers that you paste in (or the laptop extension fills in), forwards messages, and works with companies through the connector and legal requests.

**How does the protected email reach my real inbox?** Emails arrive at TraceErase's server, the sender is checked, and good emails are forwarded to your real email. TraceErase does not need to open your personal Gmail for this. Email content is never stored.

**Why not make protection fully automatic?** It is almost automatic: you set it up once. Phones and browsers require the user to choose what goes into another app's form, some sign-ups (banks, government, UPI) need your real details, and privacy law requires your consent.

**How are apps I already signed up for handled?** TraceErase finds them (Gmail scan, "Sign in with Google" list, manual add), helps you switch them to protected emails, sends legal access requests, and then sends deletion requests. Leaks from before the switch cannot

be proven by the canary.

**Is the canary email the only way to know where data went?** No. There are six ways: canary email, canary phone number, company receipts from the connector, legal data-access requests, privacy policies and Data Safety labels, and the user forwarding suspicious messages. No single method sees everything; together they cover most of it.

**What about SMS and WhatsApp?** The protected phone number catches calls, SMS and WhatsApp messages from unknown senders. No app can read your WhatsApp chats, but you can forward suspicious messages to TraceErase.

**Does a company need hardware? Who pays?** No hardware. The connector is free, open-source software. Blockchain writes are bundled hourly, so the cost is almost zero.

**How do we know a company uses the connector?** Connected companies appear as connected in the app and send receipts. For everyone else, TraceErase works in non-connected mode.

**Why would companies agree to reveal sharing details?** The law already requires them to delete on request and tell users who received their data. The connector saves them money, proves compliance, and only the user sees their own trail.

**What details are in each record?** User code (not the name), from and to, data categories, purpose, legal basis, retention period, time, data fingerprint and company signature. Readable details stay in the encrypted database; only fingerprints go on the blockchain.

**How does AI read a company's terms?** Privacy policies are public and legally required. AI reads them from the Play Store or App Store listing, the Data Safety or privacy labels, and the company website. If a policy is vague, AI sends a legal request asking the company to name its partners.

**Who decides how long data is kept, and how do we know it is deleted afterwards?** The company declares it, as the law requires. Connected companies sign it in receipts; for others, AI reads it from the policy or asks. TraceErase sets a timer, then expects an expiry receipt or confirmation. No confirmation means Overdue and escalation, and the canary keeps checking.

**How does TraceErase ask a non-connected company?** By email to the grievance or privacy officer, sent from your protected email for that app; through the company's own request form; or by a formal letter.

**What if an OTP expires while I am asleep?** The OTP arrives at TraceErase's server, and the AI submits it automatically, only for deletion requests you started. If it still fails, the AI requests a new one and re-sends the request.

**Is deletion done by company staff or automatically?** Connected companies delete automatically after identity and legal hold checks. Non-connected companies delete through their staff. Some data, such as tax invoices, must be kept by law and is reported as partial deletion.

**What does "claimed deleted" need to include?** What was deleted, the date, what was kept and why, which partners were informed, a reference number, and an official sender. The email's DKIM signature proves it is genuine, and its fingerprint is stored on the blockchain.

**Is the canary watching every company?** Yes, every company that has a protected email, from the day it got that email. Old real-email exposure cannot be proven.

**Is the complaint filed automatically?** The grievance to the company is sent automatically, and the complaint is prepared automatically. You submit it with one tap, or turn on auto-filing in settings.

**Does blocking a protected email block the real company too?** No. Smart blocking blocks only the leaker. If leaks keep happening, the protected email is rotated and the old one is closed.

**How is data deleted from the company that received it secretly?** TraceErase blocks it, asks where it got the data, demands deletion with the canary evidence, and includes it in the complaint.

**Who is the privacy regulator?** A government authority that enforces privacy law: the Data Protection Board of India, EU Data Protection Authorities, the UK's ICO, or the California Privacy Protection Agency. It can order deletion and impose fines.

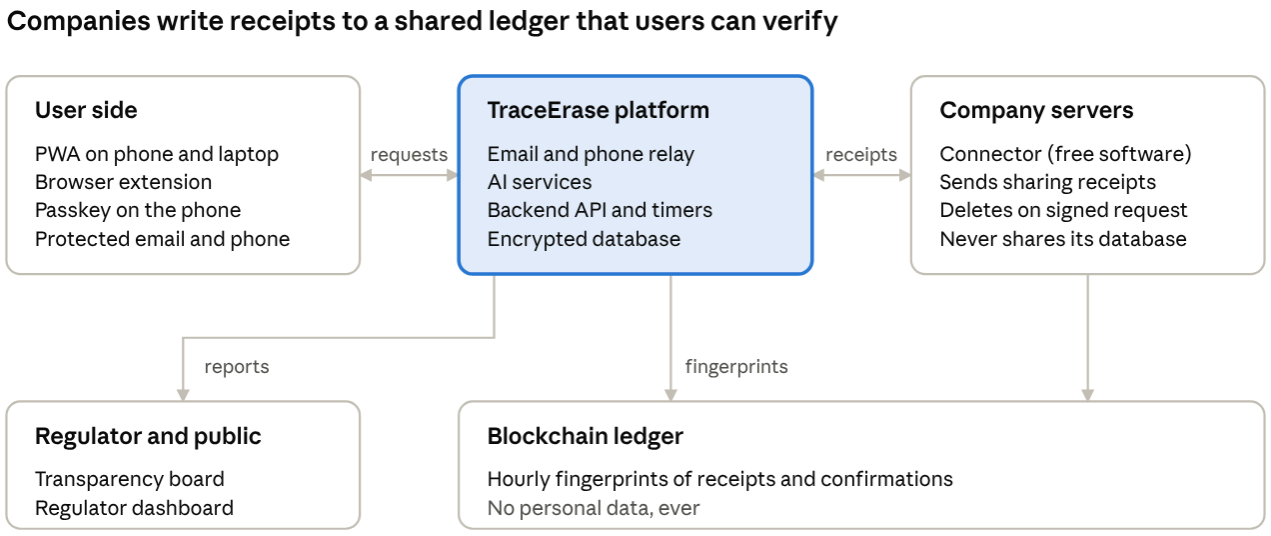
**Why is official regulator use not available yet?** The dashboard is built and works now as a public transparency board. Official use by a regulator needs a government partnership, which is paperwork, not technology.

**Will users trust TraceErase with sensitive data?** TraceErase stores only your real email and phone, never stores email content, masks personal details before AI reads anything, encrypts everything, keeps nothing personal on the blockchain, and can itself be deleted with a signed receipt.

**What happens after the certificate?** The certificate is proof. Verified and partial deletions are solved. Claimed deletions stay under canary watch. Ignored requests go to the regulator, who can order deletion and fine the company. Leakers are blocked instantly.

## 12. Technical architecture and tech stack

TraceErase has three layers: the user app, a backend with AI and the email and phone relay, and a shared ledger that companies write to through the connector.




TraceErase architecture · 5 parts

Users and companies never share databases: companies send receipts, the platform keeps readable details encrypted, and only fingerprints reach the blockchain.

## 12.1 Components

| Component | What it does |
|---|---|
| TraceErase PWA | User portal on phone and laptop: sign-ups, data map, one-click delete, dashboard, certificate |
| Browser extension | Fills protected emails into sign-up forms and company request forms on a laptop |
| Email relay | Receives mail for protected emails, checks the allowed senders list, forwards or blocks, logs the sender |
| Phone relay | Virtual number that forwards calls, SMS and WhatsApp, and logs senders |
| AI services | Discovery, policy reading, request writing, reply checking, OTP handling, leak detection, complaints |
| Backend API | Accounts, requests, timers, statuses, follow-ups, notifications |
| Encrypted database | Readable receipt details and request records, visible only to the owner |
| Smart contract on a blockchain | Stores hourly Merkle roots of receipts and confirmation fingerprints |

| Component | What it does |
|---|---|
| Connector (SDK / API) | Runs on company servers; sends sharing, deletion and expiry receipts; receives forwarded deletion requests |
| Company, regulator and public portals | Compliance view, transparency board, regulator dashboard |

## 12.2 Tech stack

| Layer | Technology |
|---|---|
| Frontend | React + Vite, installable as a PWA |
| Browser extension | Chrome extension (Manifest V3) |
| Backend | Node.js on Firebase Cloud Functions |
| Database | Cloud Firestore, encrypted fields |
| Authentication | Firebase Auth plus passkeys (WebAuthn) |
| Blockchain | Solidity smart contract on Polygon Amoy testnet, or a local Hardhat chain for the demo |
| AI models | Gemini or NVIDIA NIM for reading, writing and classification |
| Email discovery | Gmail API, read-only, with user permission |
| Email relay | Inbound email webhook for protected addresses (simulated in the demo) |
| Phone relay | Virtual number provider (simulated in the demo) |
| Notifications | Web push notifications for the PWA |

## 12.3 Smart contract functions

- `anchorBatch(merkleRoot, batchTime)`: writes one combined fingerprint for an hour of receipts.
- `recordShare(userHash, fromCompany, toCompany, dataHash, retentionUntil)`: the logical receipt, included in a batch.
- `requestDeletion(userHash, requestId, userSignature)`: records that a signed deletion request was made.

- `confirmDeletion(requestId, company, keptReason, companySignature)`: a connected company's signed deletion receipt.
- `recordConfirmationEmail(requestId, company, emailHash)`: the fingerprint of a DKIM-verified confirmation from a non-connected company.
- `getStatus(requestId)`: returns verified, partial, claimed, pending, no response or overdue for each company.

### 12.4 Data flow in one line each

1. **Sign-up:** user gets a protected email, pastes it into the app, and TraceErase records the app as its owner.
2. **Sharing:** a connected company's system calls the connector, which sends a receipt; the details go to the database and the fingerprint into the hourly batch.
3. **Deletion:** the user signs one request; connected companies delete and send receipts; the AI handles everyone else by email, form or letter.
4. **Monitoring:** the email and phone relays check every sender against the allowed list; unknown senders trigger leak alerts.
5. **Escalation:** timers drive reminders, grievances, complaints and the transparency board.

## 13. Demo plan and measured improvement

The organisers ask teams to show the security failure, show how the solution fixes it, and measure the improvement; the demo is built in exactly those three parts.

### 13.1 Part 1: show the problem (without TraceErase)

1. Sign up on a demo shopping app that shares data with four simulated partner services.
2. Delete the account. The app says "Deleted".
3. Open the partner services and show that three of the four still hold the data.

### 13.2 Part 2: show the solution (with TraceErase)

1. Sign up again with a protected email. Show the sharing receipts appearing live, with fingerprints on the blockchain.
2. Tap "Delete my data everywhere" and confirm with a fingerprint.
3. Watch signed receipts arrive from connected services; watch the AI send a legal request to a non-connected one and auto-handle its OTP.
4. One partner ignores the request: show the reminder, grievance and prepared complaint.

5. A marketing email arrives at the protected email from an unknown company. Show the leak alert, smart blocking and the evidence pack.
6. Download the deletion certificate and verify a receipt against the blockchain.

### 13.3 Part 3: show the numbers

| Metric | Without TraceErase | With TraceErase |
|---|---|---|
| Copies of data found | 1 of 5 | 5 of 5 |
| Copies confirmed deleted | Unknown | 4 confirmed, 1 flagged |
| Proof of deletion | None | Signed, tamper-proof receipts |
| Time to detect non-compliance | Never | At the deadline, or instantly via the canary |
| Effort for the user | Contact each company separately | One click |

The figures above are for the demo scenario and will be measured live during the presentation.

## 14. Honest limitations and future scope

Stating limits openly makes the project more credible to judges; each limit below has a mitigation or a next step.

### 14.1 Limitations

| Limitation | How TraceErase handles it |
|---|---|
| Cannot see inside company databases | Relies on signed receipts, legal requests, timers and canary checks |
| Leaks before an app got a protected email cannot be proven | Legal access requests for old accounts; protection is strongest from day one |
| Banks, UPI and government services need the real phone number | Real number used for regulated services only |
| Non-connected companies give weaker proof | DKIM-verified confirmations fingerprinted on the blockchain, plus canary monitoring |

| Limitation | How TraceErase handles it |
|---|---|
| WhatsApp chats cannot be read | The protected number catches unknown senders; users can forward messages |
| Official regulator use needs a partnership | The dashboard works now as a public transparency board |
| In the demo, partner companies are simulated | Shows exactly how Mode A works end to end |
| Data used to train AI models is hard to delete | Flagged as an open item; see future scope |

### 14.2 Future scope

- Registration as a DPDP Consent Manager with the Data Protection Board.
- Direct integration with regulator complaint systems.
- Machine unlearning checks for data used to train AI models.
- Support for more privacy laws and languages, with voice access for low-literacy users.
- Data broker search: finding companies that sell personal data and sending them deletion requests.
- Industry adoption through a "TraceErase Verified" programme.

## 15. 24-hour build plan and summary

A team of six can build a working demo in 24 hours by splitting the system into six parallel parts.

### 15.1 Build plan

| Member | Builds |
|---|---|
| 1 | Smart contract (Solidity), hourly batching and blockchain connection |
| 2 | Demo ecosystem: a shopping app plus four simulated partner services, two of them with the connector |
| 3 | User PWA: sign-up helper, data map, one-click delete, dashboard, certificate |

| Member | Builds |
|---|---|
| 4 | AI services: discovery, policy reader, legal request writer, reply checker, auto-OTP |
| 5 | Email and phone relay (simulated), canary leak detector, smart blocking, company and regulator portals |
| 6 | PPT, demo script, testing and the measured numbers |

## 15.2 Every part in one line

| Part | Job |
|---|---|
| PWA | One web app on phone and laptop |
| Browser extension | Fills protected emails on laptop sign-ups and company forms |
| Passkey | Proves requests are really from the user |
| Protected email and phone | Hide the real identity, forward messages, act as canaries |
| Connector | Free software that sends receipts when a company shares or deletes data |
| Blockchain | Unchangeable fingerprints, bundled hourly, nearly free |
| Encrypted database | Full details, visible only to the user |
| AI | Finds old apps, reads policies, writes requests, handles OTPs, checks replies, detects leaks, builds complaints |
| Transparency board and regulator dashboard | Public accountability now, official use later |

## 15.3 Summary

TraceErase gives every person a protected identity, a map of where their data went, one-click deletion everywhere, and proof of who obeyed and who lied, whether the company cooperates or not.

**Pitch line:** "Apps say 'deleted'. TraceErase proves it, everywhere your data went."